SUN CHALA LE SAHI HAI POORA CODE OR KUCH KARNA HO TO BATA DE

que -1

Poisson Distribution Problem

```
lambda <- 3
```

```
p0 <- dpois(0, lambda) p3 <- dpois(3, lambda) p_le_4
<- ppois(4, lambda)
```

```
cat("Poisson Results:\n") cat("P(X = 0) =", p0, "\n")
cat("P(X = 3) =", p3, "\n") cat("P(X <= 4) =", p_le_4, "\n")
```

Que-2

Two-Proportion Hypothesis Test

Data

```
x <- c(480, 420) n <- c(1000, 800)
```

Manual Z-test

```
p1 <- x[1] / n[1] p2 <- x[2] / n[2]

p_pool <- sum(x) / sum(n)

se <- sqrt(p_pool * (1 - p_pool) * (1/n[1] + 1/n[2]))

z <- (p1 - p2) / se

z_crit <- qnorm(0.995)

cat("Manual Z-Test:\n") cat("Z =", z, "\n") cat("Critical Z =", z_crit, "\n")

if (abs(z) > z_crit) { cat("Decision: Reject H0\n\n") } else { cat("Decision: Fail to Reject H0\n\n") }
```

p-value approach

```
test <- prop.test(x, n, correct = FALSE)

cat("R Test Output:\n") print(test)

if (test$p.value < 0.01) { cat("Decision: Reject H0\n") } else { cat("Decision: Fail to Reject H0\n") }
```